

Linear-Exponential Algorithm-based Open Source Dive Computer

Sebastian Thomas

Bachelor Thesis

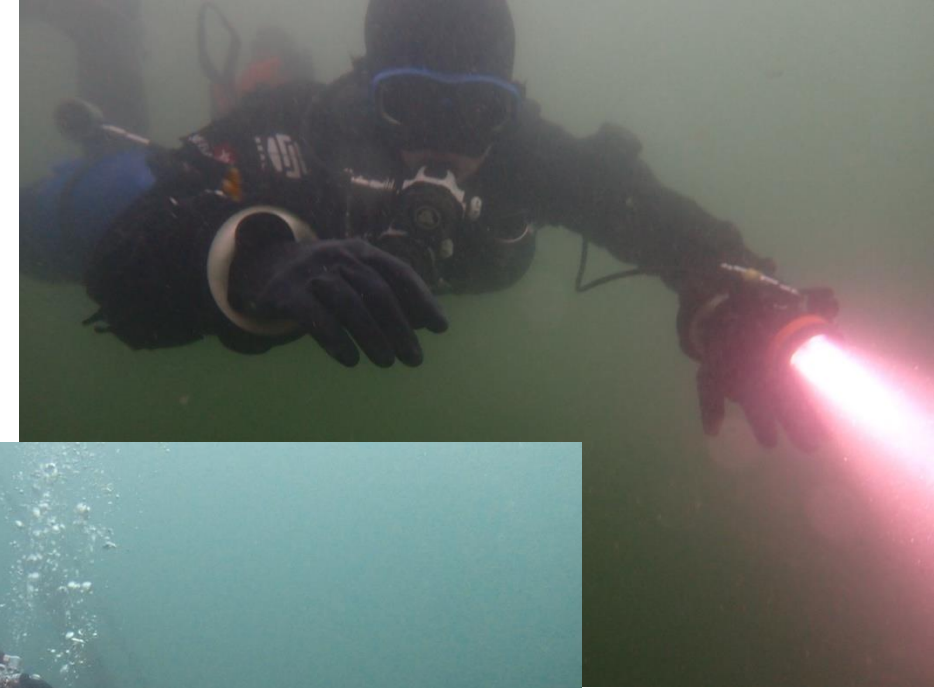
Supervisor: Tommaso Polonelli

Professor: Prof. Luca Benini

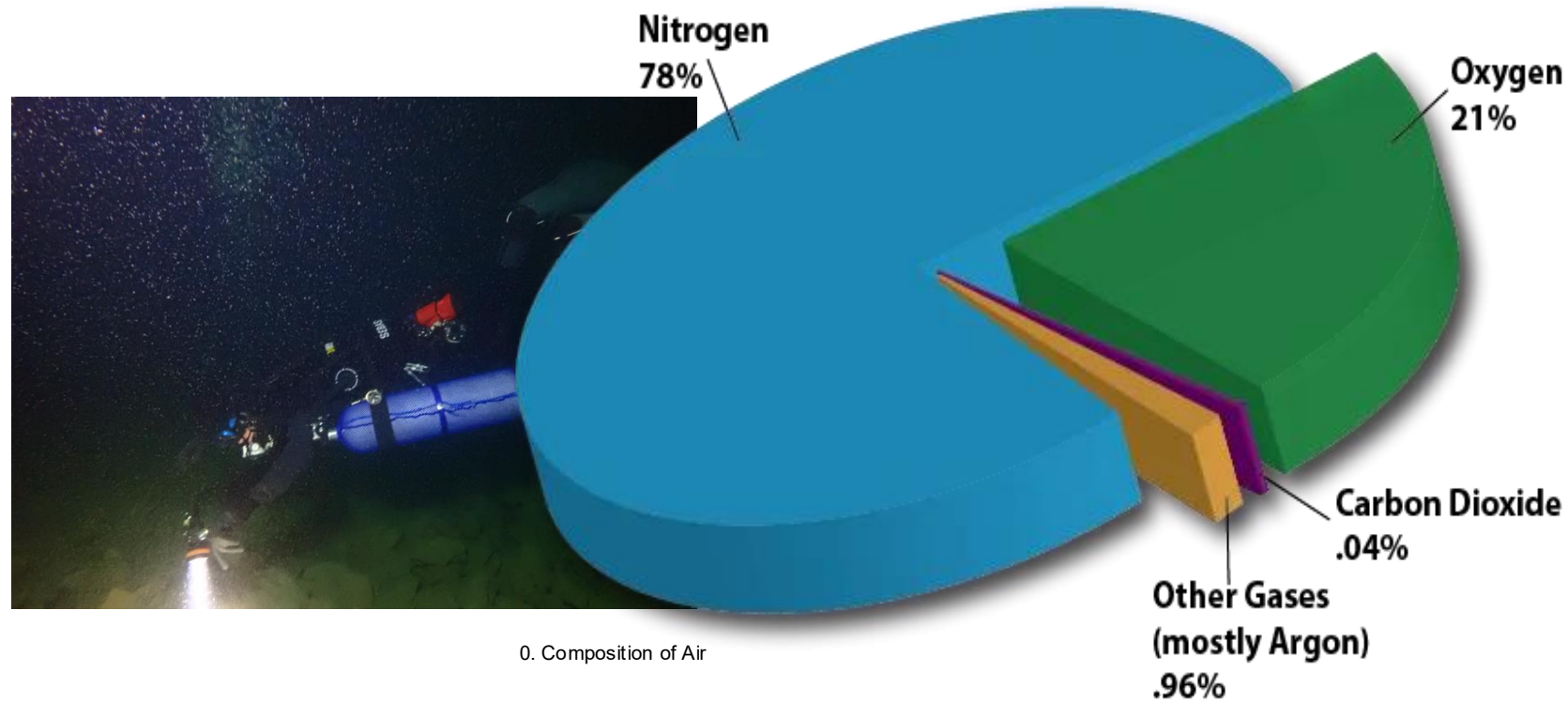
16. June 2026



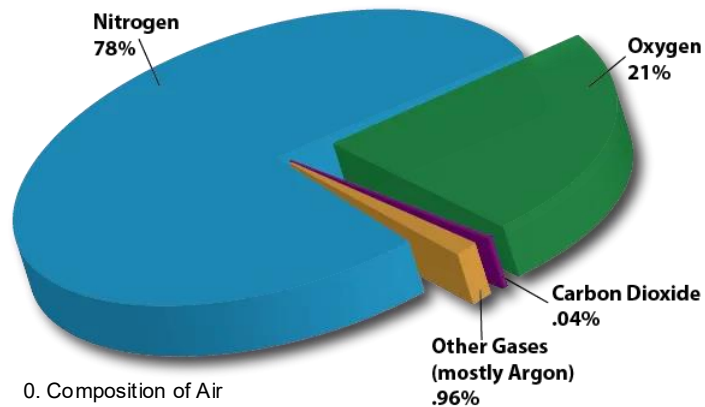
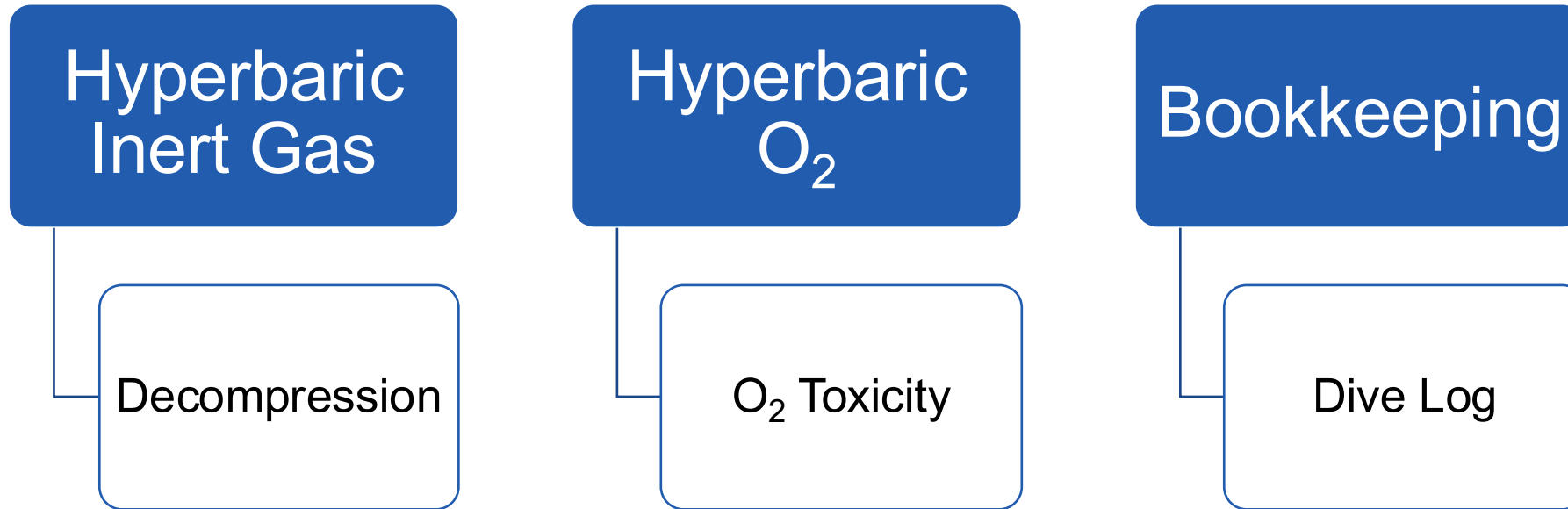
Introduction: Diving & Hyperbaric Environments



Introduction: Diving & Hyperbaric Environments



Introduction: Dive Computers



Introduction: Dive Computers

Hyperbaric
Inert Gas

Decompression

Hyperbaric
O₂

O₂ Toxicity

Bookkeeping

Dive Log



3. Shearwater Perdix 3 (2026)



2. Mares Sirius (2023)



1. Hans Hass DECO-BRAIN ® (1980s)

Problem Description



COMMERCIAL SETTING

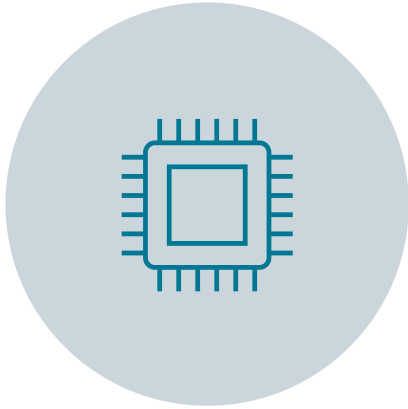


FIXED DECOMPRESSION
ALGORITHMS



FIXED LOG RATE

Thesis Goals



DESIGN & BUILD COMPACT
PCB

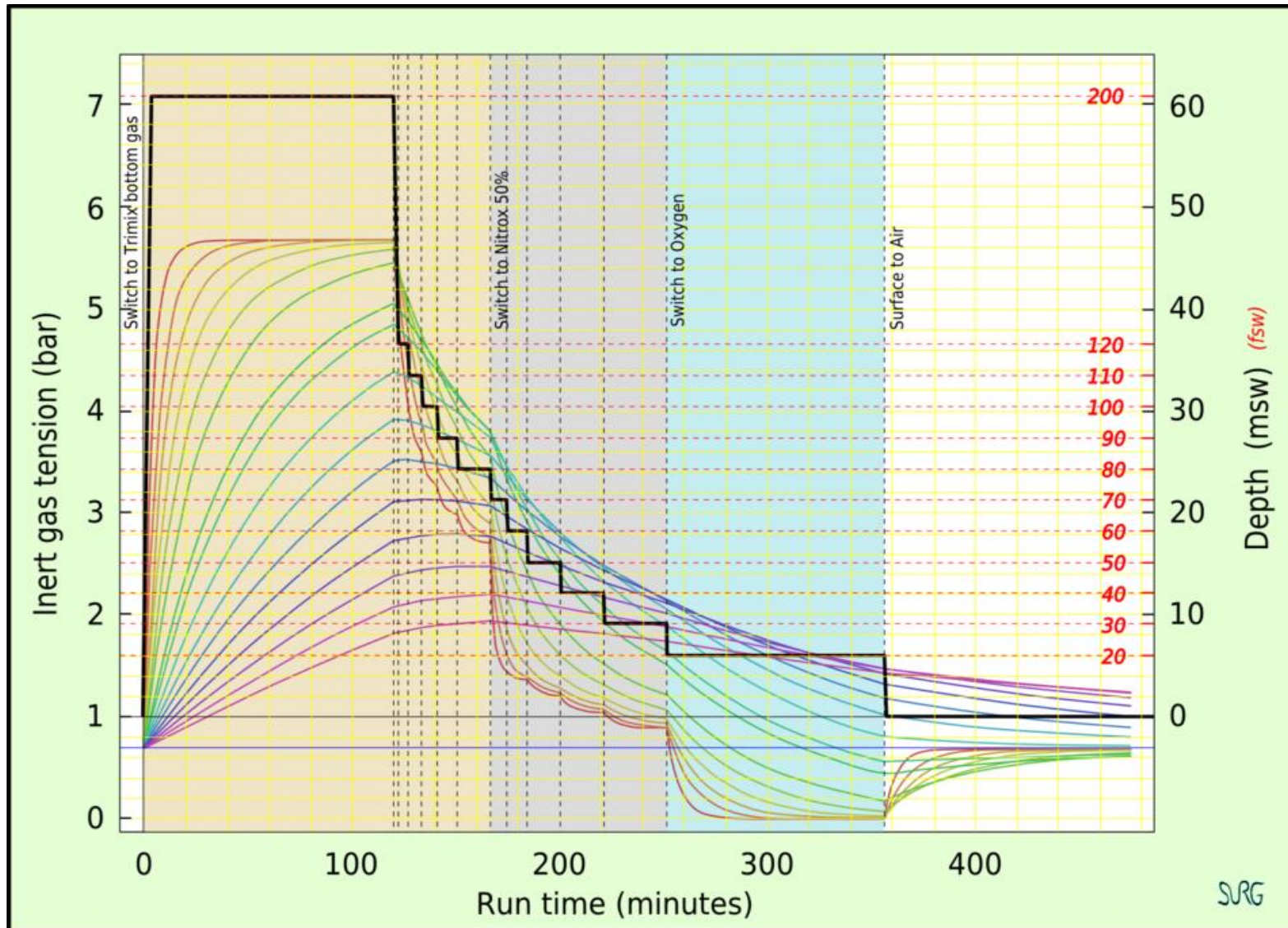


IMPLEMENT & PROFILE
LINEAR-EXPONENTIAL
DECO-ALGORITHM



IMPLEMENT & PROFILE
VARIABLE-RATE
ALGORITHMS

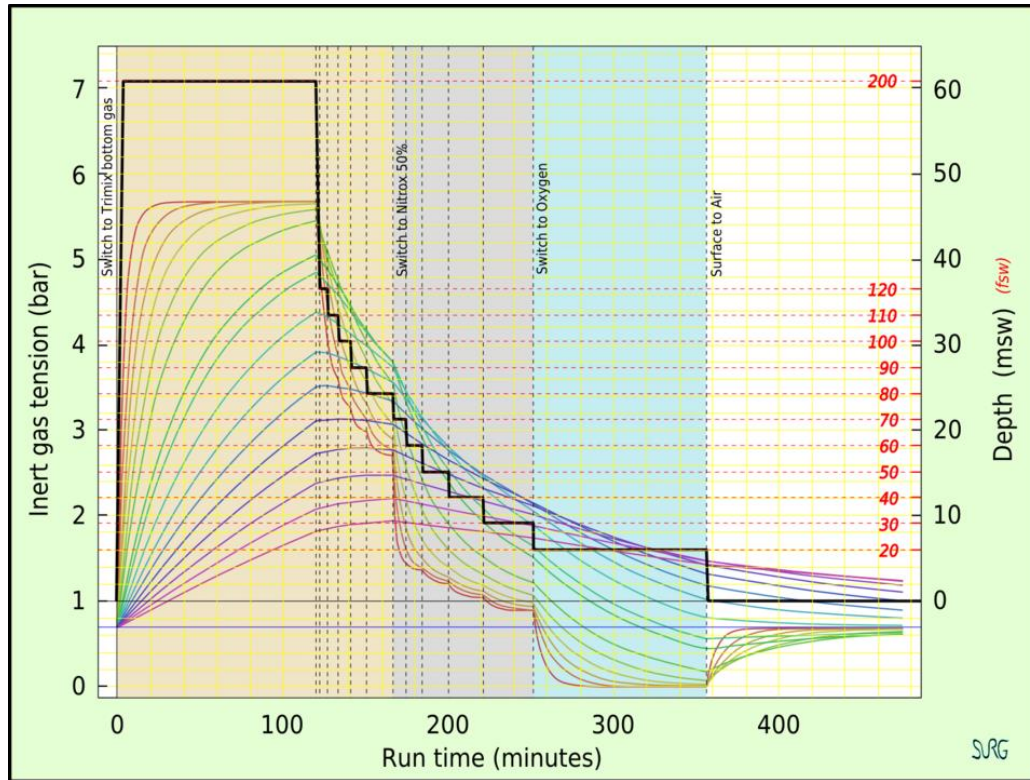
Background: Decompression Algorithms



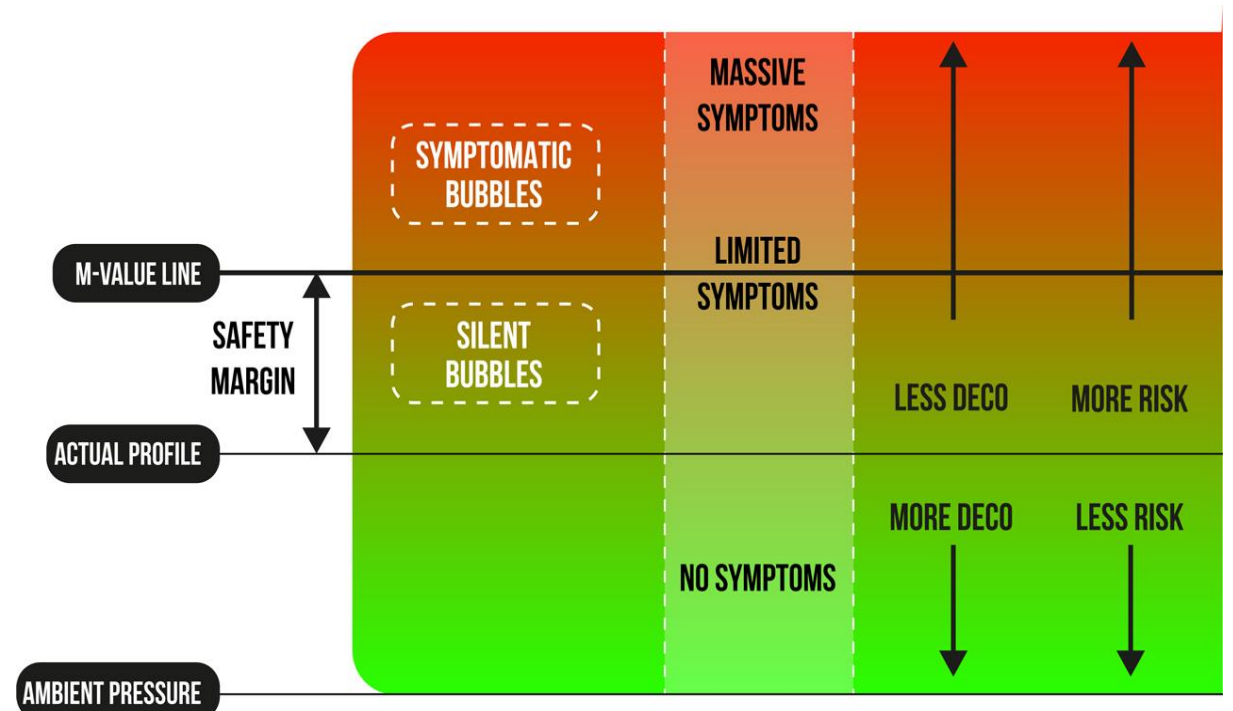
5. Example Decompression Model

5. https://en.wikipedia.org/wiki/File:Inert_gas_tensions_in_the_tissue_compartments_during_a_decompression_dive.png

Background: Hyperbaric Environments



5. Example Decompression Model



4. The M-Value

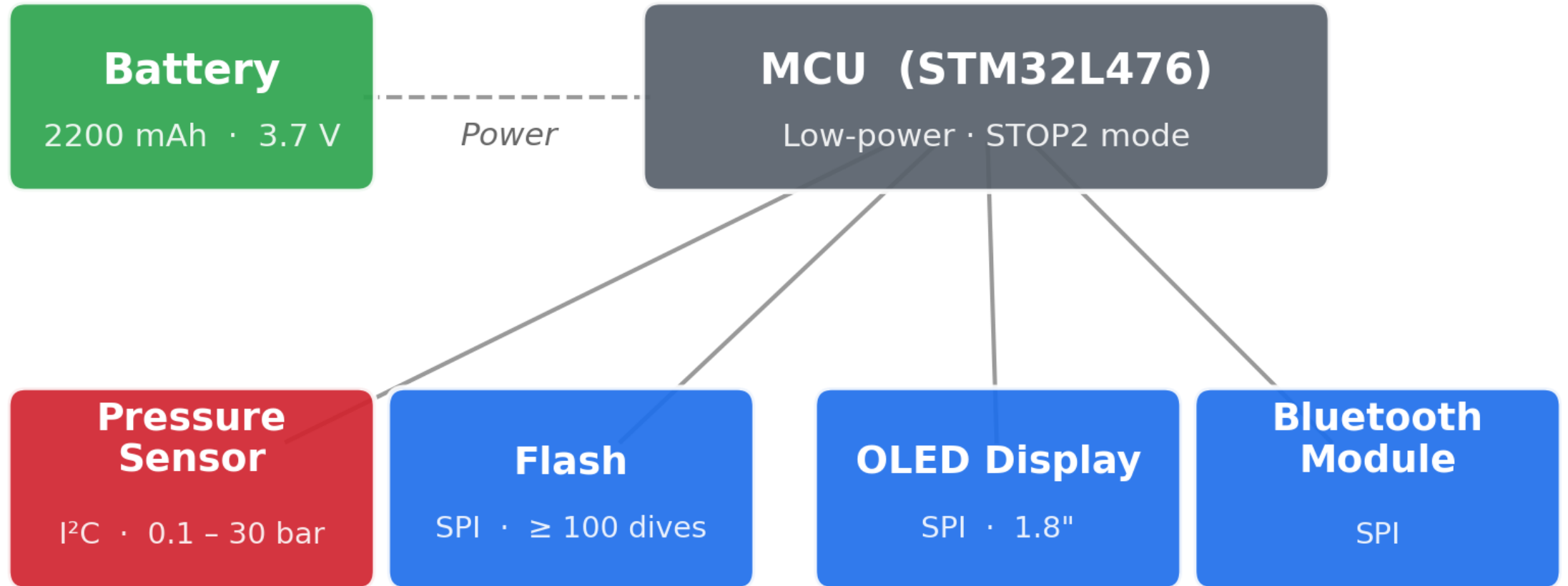
4. https://www.tdisdi.com/wp-content/uploads/2022/08/M_Value_Graph-1.jpg

5. https://en.wikipedia.org/wiki/File:Inert_gas_tensions_in_the_tissue_compartments_during_a_decompression_dive.png

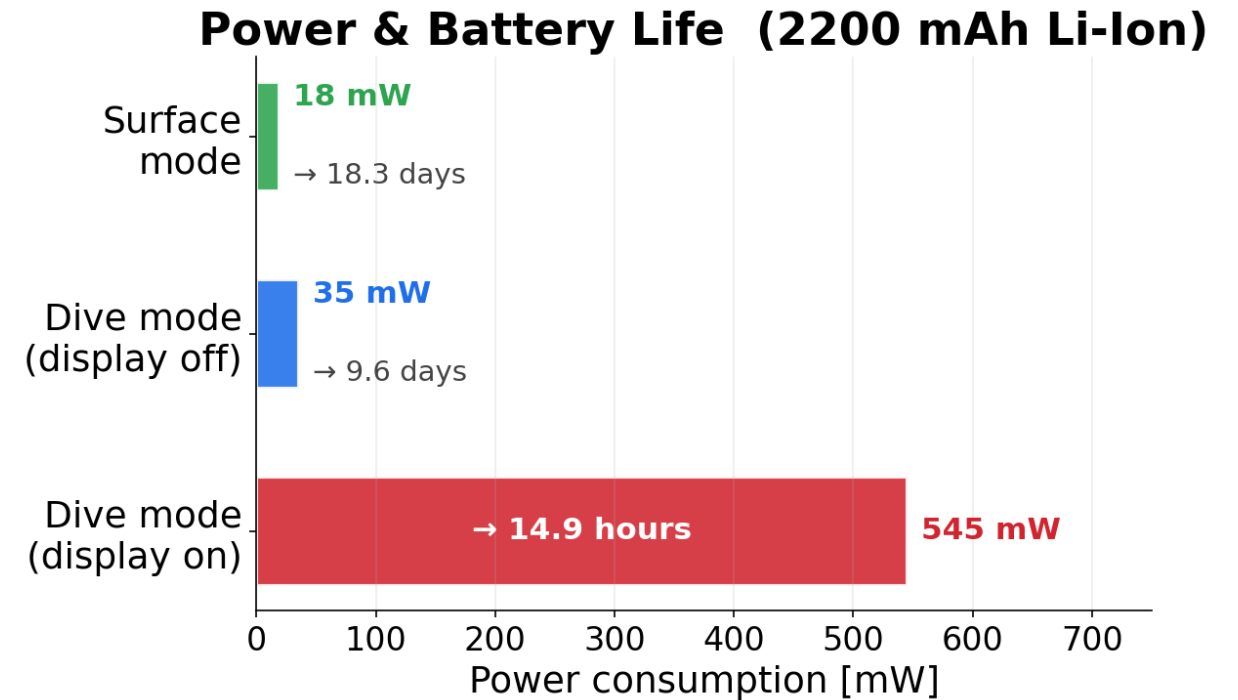
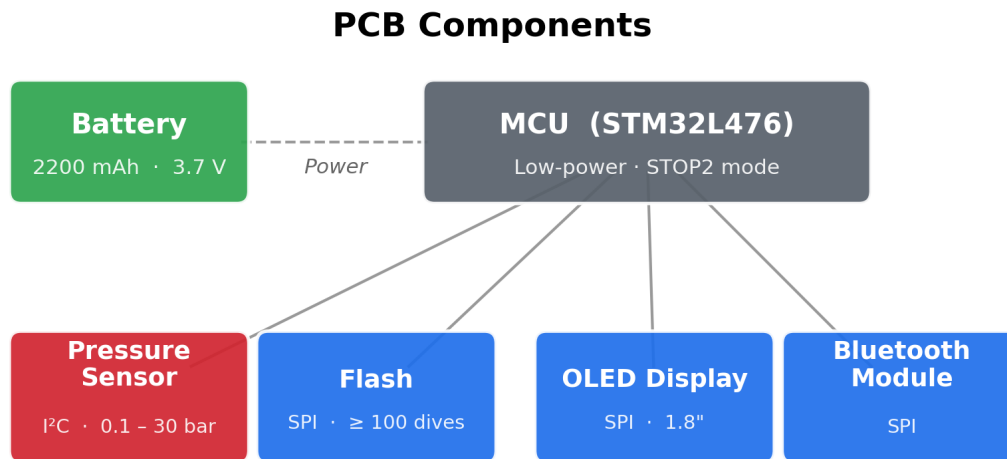
Implementation

Implementation: Hardware

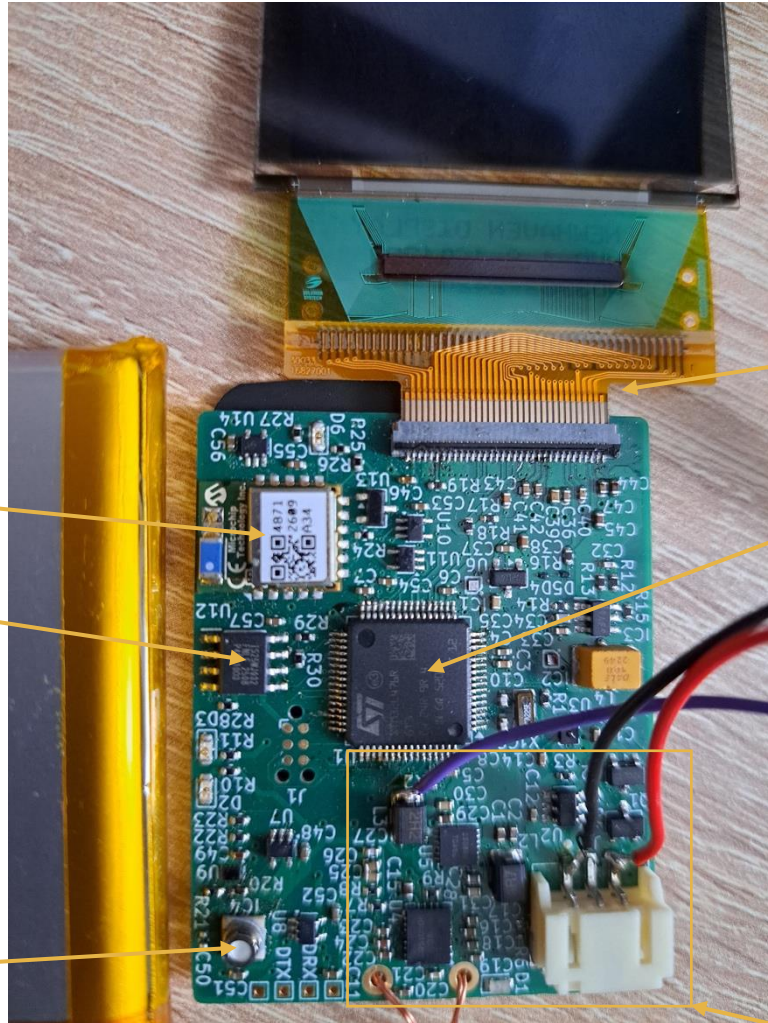
PCB Components



Implementation: Hardware

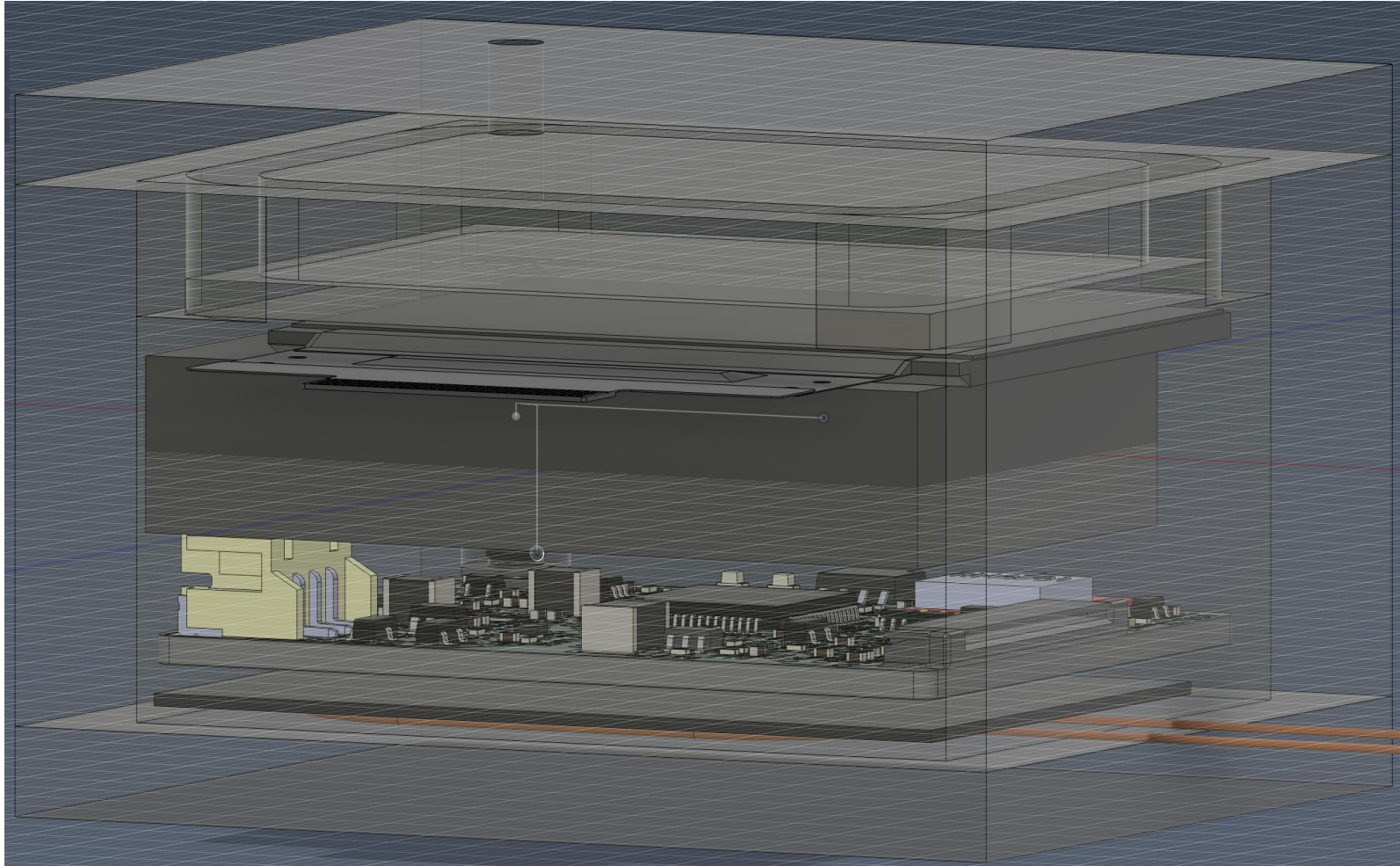


Implementation: Hardware

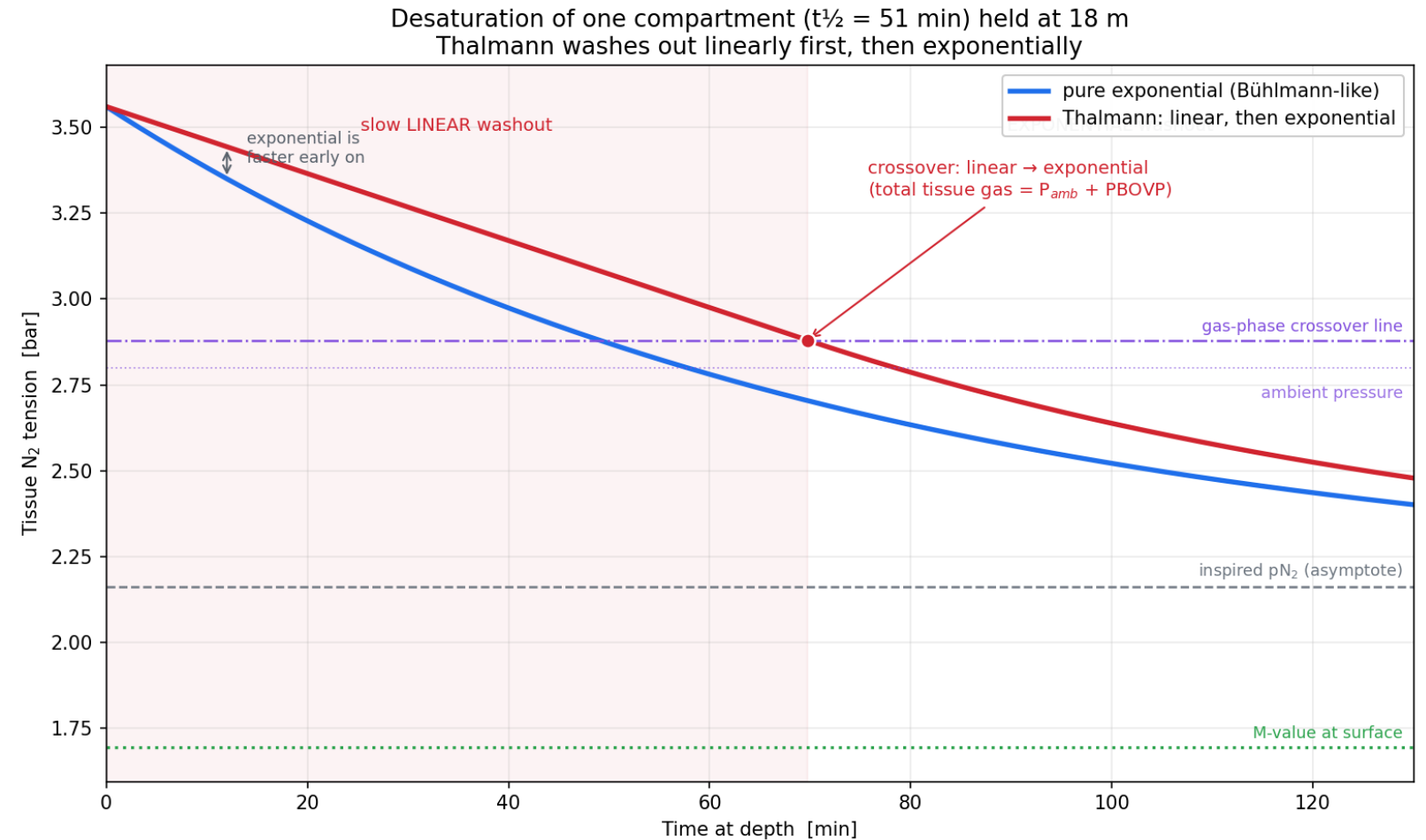


- MCU: Low-Power
- Battery: 18 days / 15 dive hours
- Pressure sensor: 0.1-30 bar
- External Bluetooth module
- Large, bright, OLED display
- Flash for >100 dives

Example: Acrylic Enclosure



Implementation: Linear-Exponential decompression algorithm



Implementation: Variable-rate algorithms

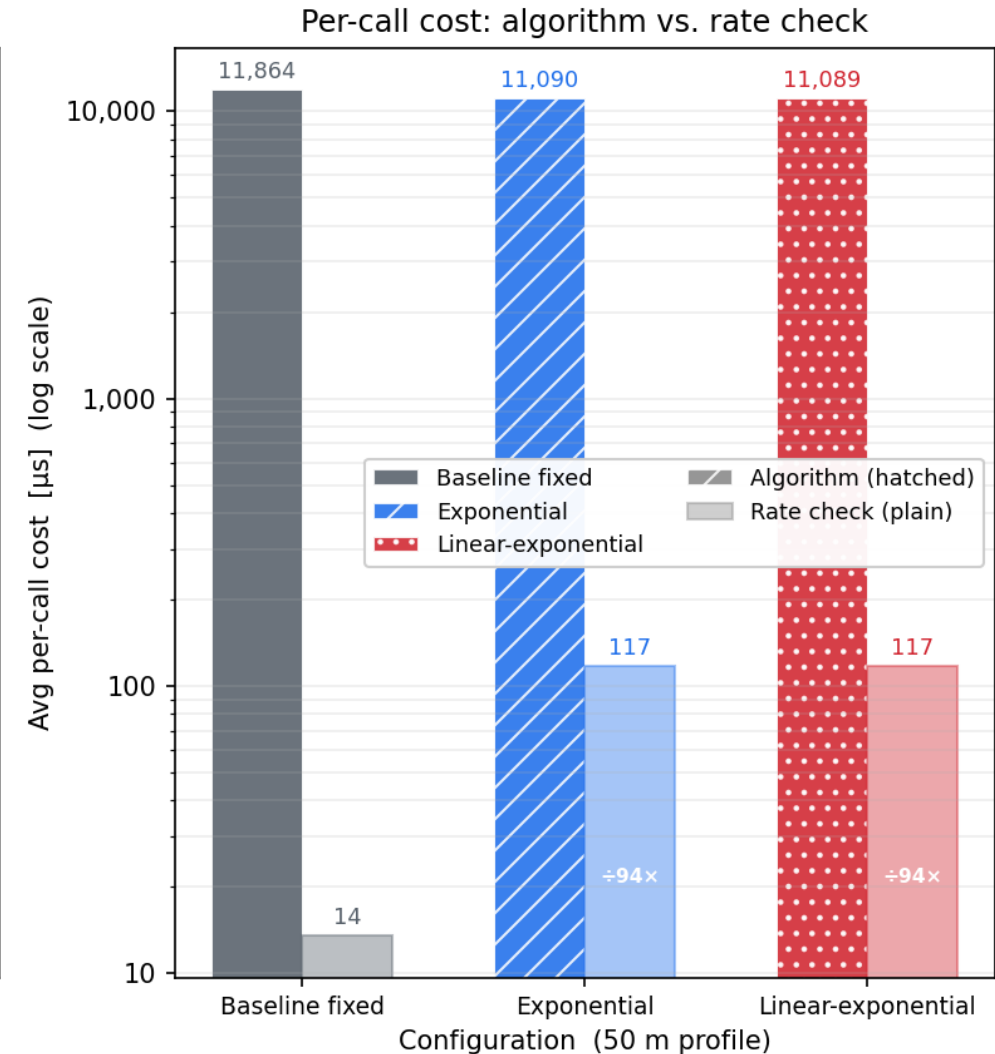
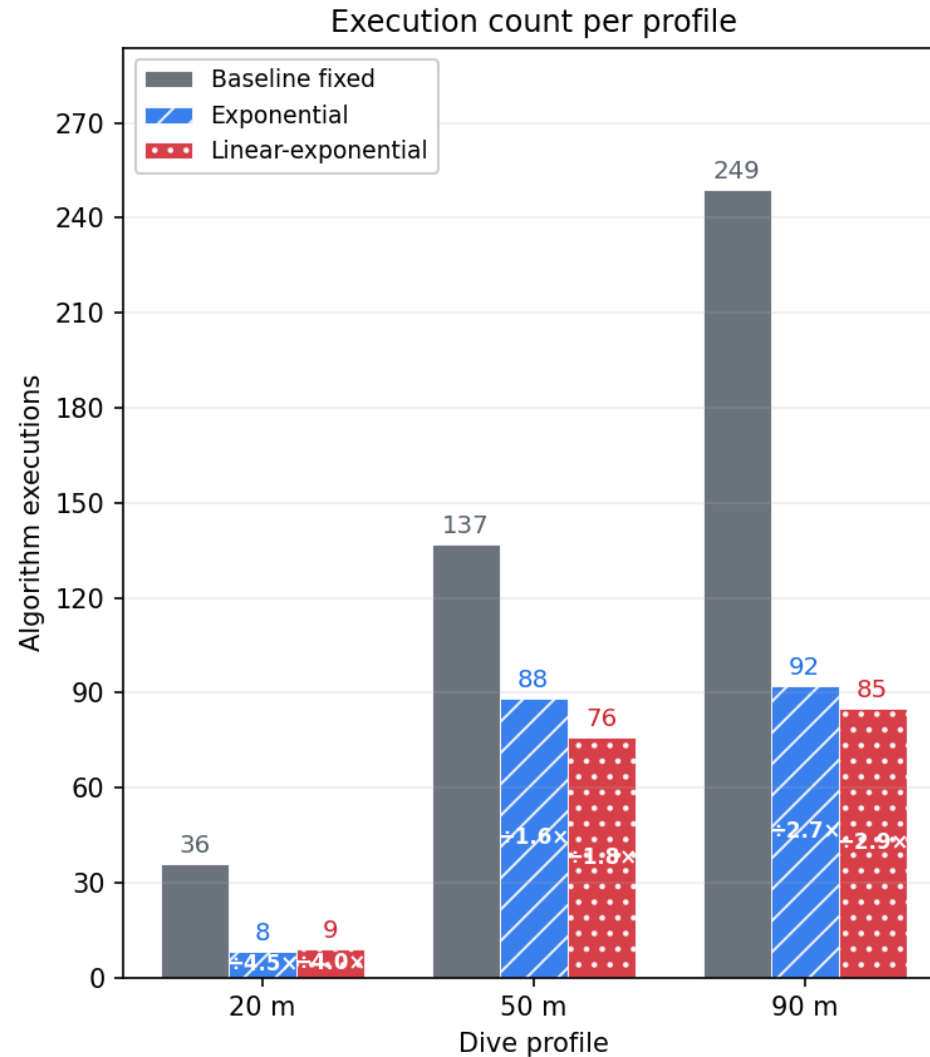
- Additive Increase (AI), Multiplicative Decrease (MD – **change**)

	DiffAimdRateAlgorithm	DynamicDiffAimdRateAlgorithm
change	$\frac{\delta v}{\delta t}$	$\frac{\delta v}{\delta t}$ or <i>fixed threshold</i>
Cycles	~1730	~1850

Results & Discussion

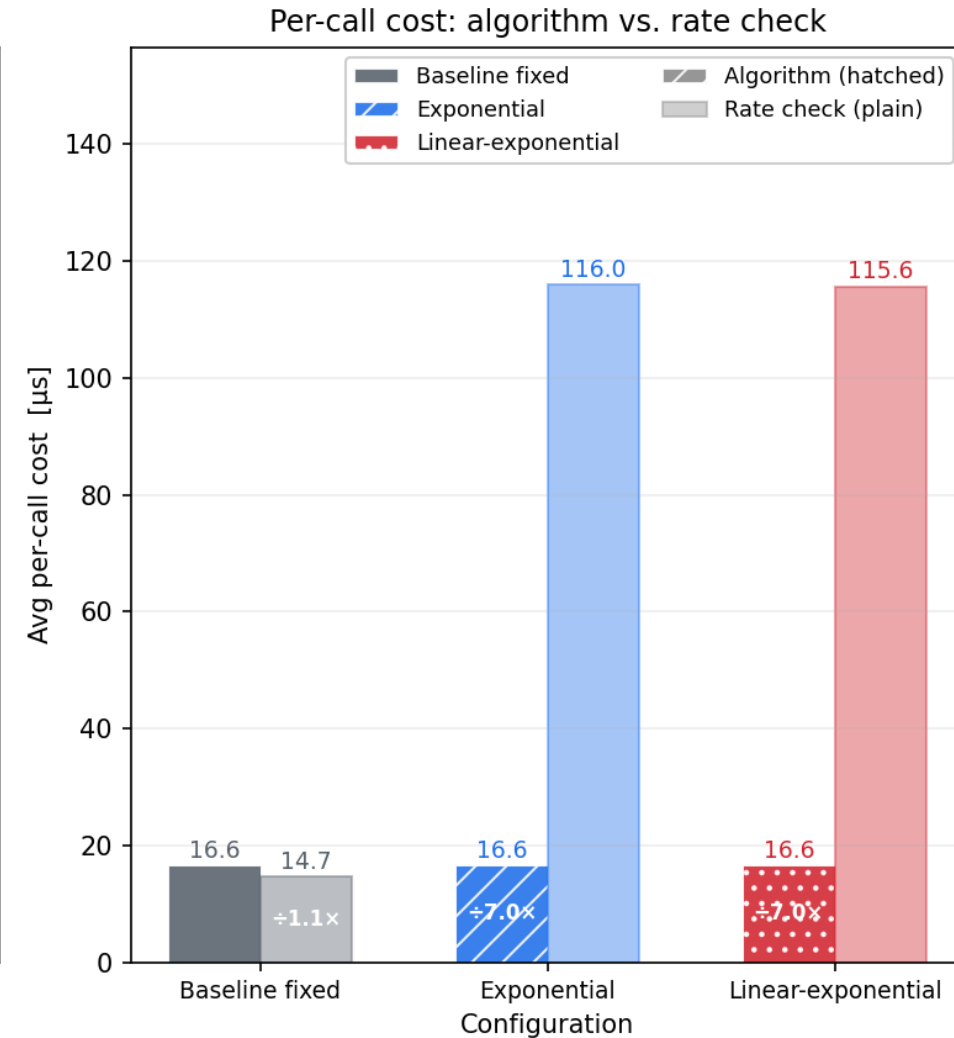
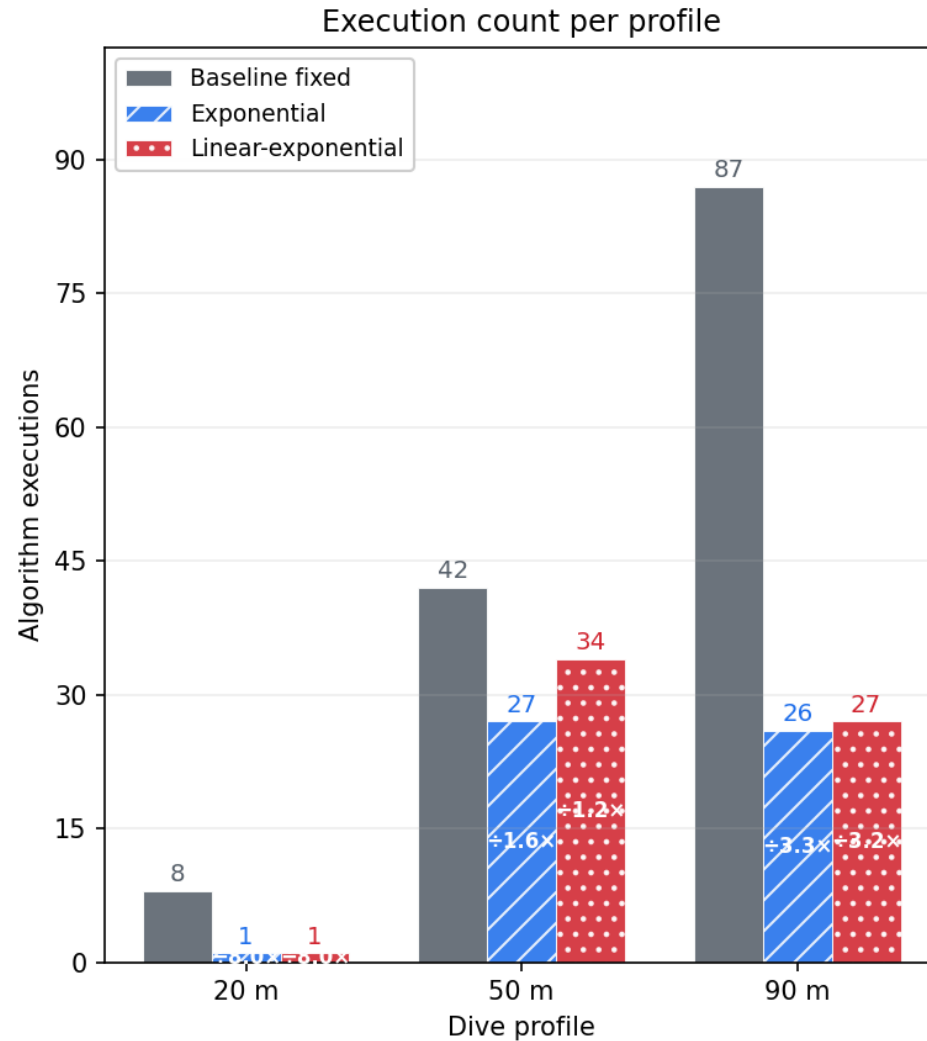
Results: Variable-rate algorithms – decompression models

Variable-rate deco scheduling: execution count & per-call cost
White labels (left) show reduction factor vs. fixed-rate baseline

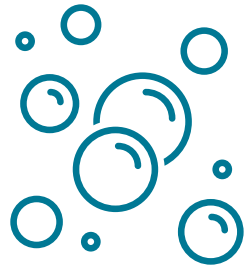


Results: Variable-rate algorithms – O2Tox calculation

Variable-rate O₂-toxicity scheduling: execution count & per-call cost
White labels (left) show reduction factor vs. fixed-rate baseline

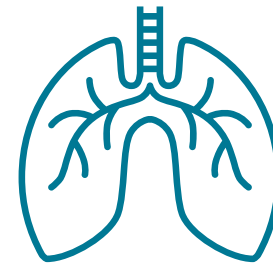


Discussion: Variable-rate for diving algorithms



Variable-rate Decompression Model

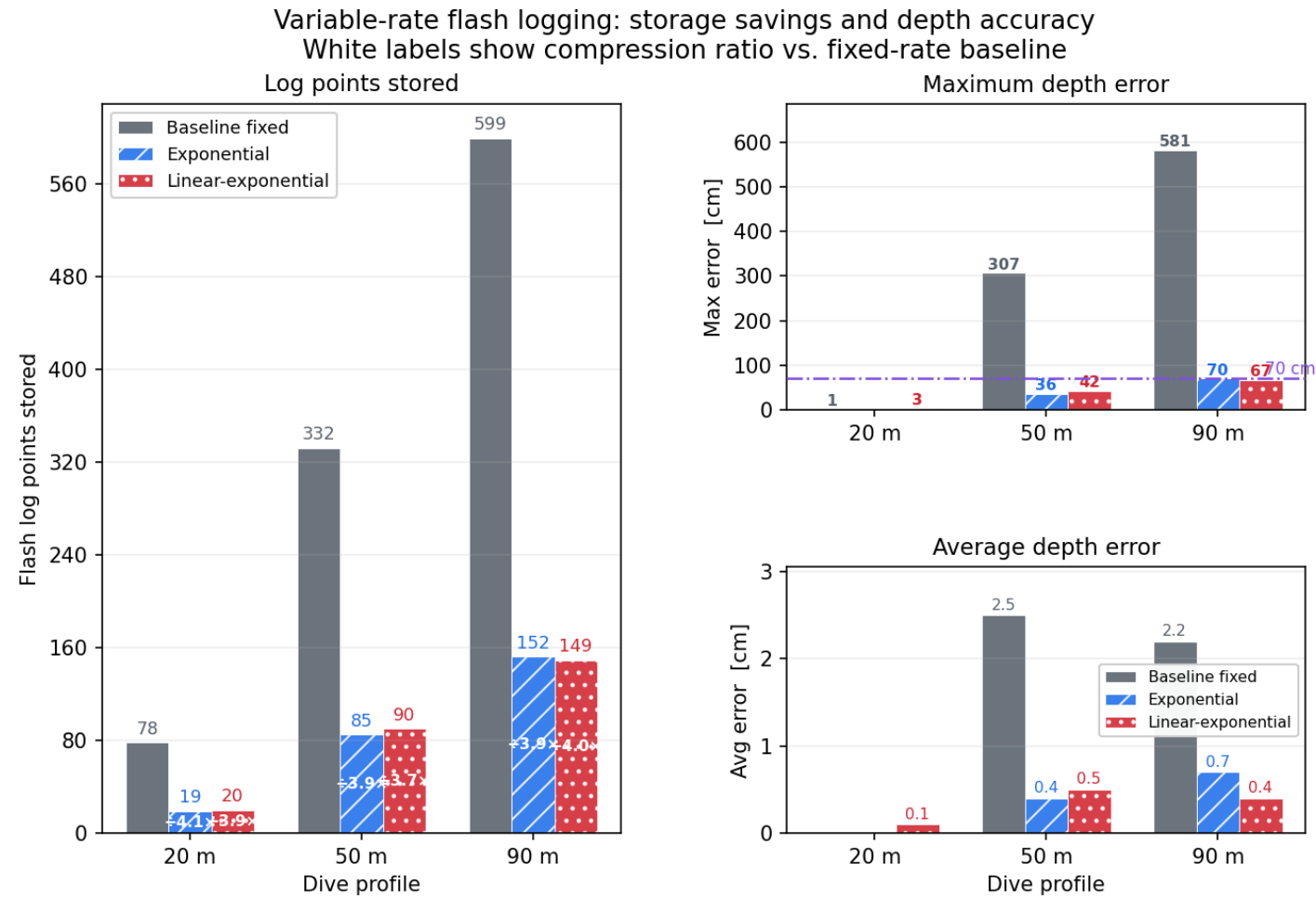
$1/3 \times$ energy required
minor/short-term deviations



Variable-rate O2Tox

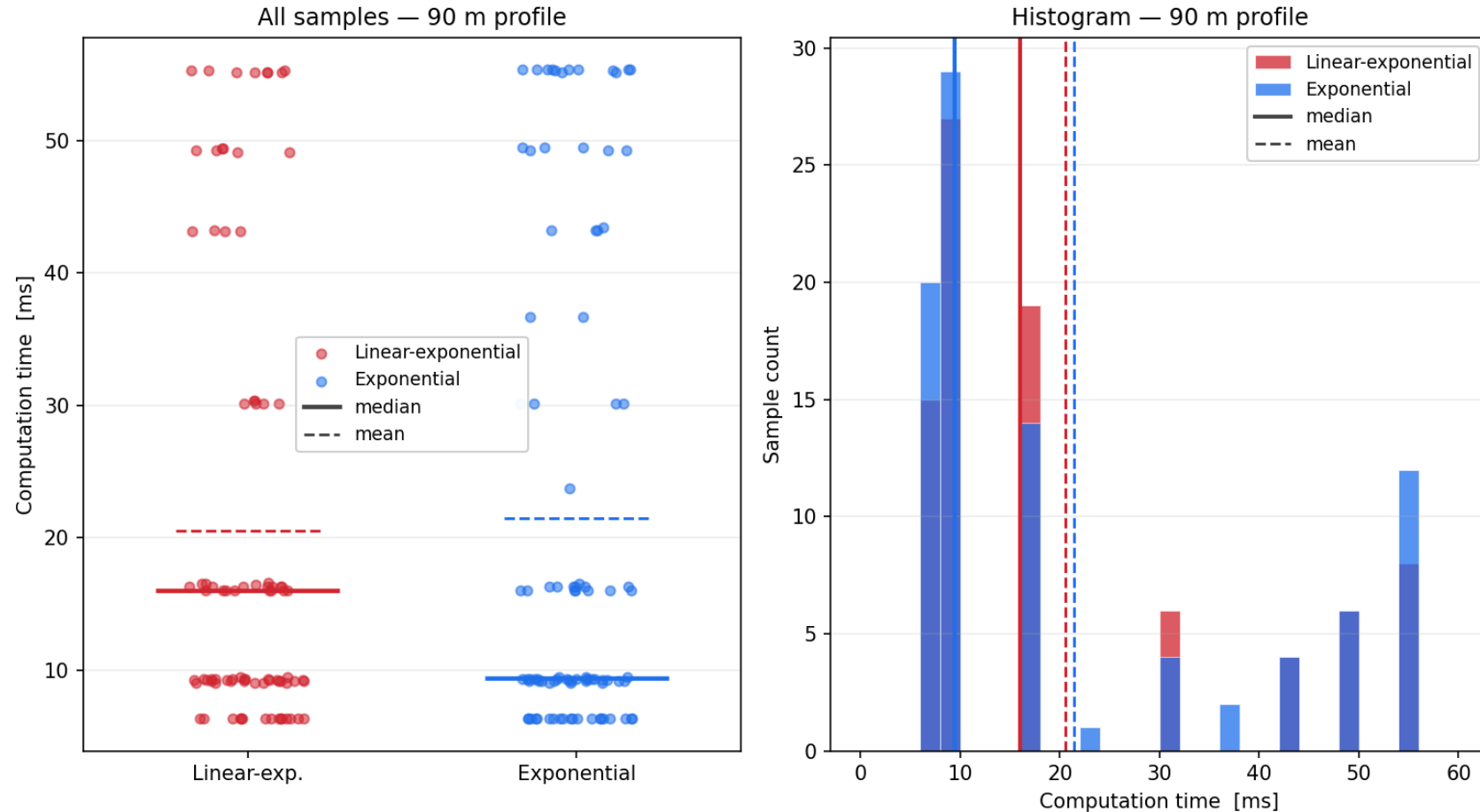
overhead too large

Results: Variable-rate algorithms – Dive Log to Flash



Results: Linear-exponential decomposition algorithm

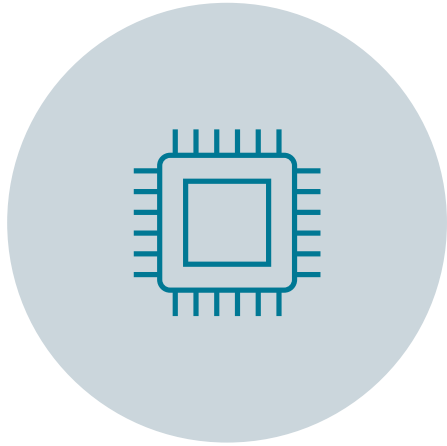
Decompression schedule computation time — 90 m dive (10 min)
STM32L476 @ 16 MHz, DWT cycle benchmarks | n=85 lin-exp, n=92 exp



Discussion: Linear-exponential algorithms & Variable-rate Flash logging

- Linear-exponential decompression algorithm
 - As expected, customizability
- Variable-rate Dive Log to Flash
 - Reduction: 3.6 ×-4.1 ×

Conclusion



WEARABLE, COMPACT,
OPEN SOURCE
HARDWARE



LINEAR-EXPONENTIAL
DECO:
 $\leq 20\%$ OVERHEAD



VARIABLE-RATE:
 $\leq 0.28 \times \text{FLASH SPACE}$
 $\leq 0.32 \times \text{DECO ENERGY}$

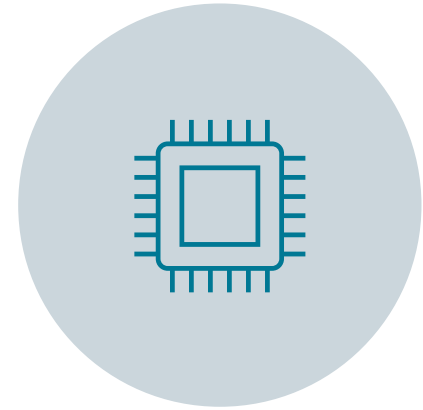
Future Work



REAL-WORLD
TESTING



OPTIMIZE
ALGORITHMS



FIX HARDWARE
PROBLEMS

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